SYSTEMS ARCHITECTURE & WORKFLOW CONCEPT

Autonomous Maintenance Request Triage & Vendor Coordination

A non-disruptive operational overlay that turns unstructured tenant messages into organized, manager-approved vendor actions.

PREPARED FOR:	Highland Property Group
ATTENTION:	Marcus Vance, VP of Operations
KNOWN PM STACK:	AppFolio / Existing PM System
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Why this concept was prepared for Highland Property Group: Based on your portfolio of scattered-site residential units, managing 24/7 tenant maintenance requests across email, portal, and text typically consumes dozens of coordinator hours each week—creating a constant tension between after-hours burnout and the risk of delayed habitability emergencies.

The Operational Friction in High-Volume Maintenance Intake

Invisible Cognitive Load

Maintenance requests arrive undifferentiated around the clock. Coordinators spend hours deciphering rambling emails, re-keying details into work orders, chasing tenants for missing access/pet details, and assessing urgency manually—while true emergencies risk sitting unnoticed until morning.

Unstructured Messages
Tenants report issues with missing details, rambling descriptions, or misleading urgency flags.

The Two-Sided Risk
Under-responding turns small leaks into $10k claims; over-responding burns budgets on weekend callouts.

After-Hours Coverage Gap
Standard business hours cover 40 of 168 weekly hours; unmonitored nights leave operators exposed.

TARGET OPERATIONAL TRANSFORMATION

ADVISORY-ONLY ARCHITECTURE

Zero emergency maintenance reports sitting unread outside office hours, with 70%+ of routine triage, clarification, and dispatch preparation handled before human review.

Instant Risk Flagging
Dangerous language (gas, flooding, electrical) is caught in seconds by escalate-only safety rules.

Hands-Off Clarification
Missing access permissions, pet status, and symptoms are gathered directly from the resident first.

One-Tap Manager Approval
Staff review prepared vendor choices and approve dispatch with a single tap. Zero auto-dispatch.

01 CURRENT FRICTION — MANUAL

- Unsorted email/portal inbox
- Staff manually extracts trade & unit
- Multiple emails to get pet/access info
- Delayed emergency phone alerting

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02 AUTOMATION OVERLAY — SYSTEM

- Instant message normalization & dedupe
- Strict JSON AI extraction of urgency & trade
- Binding escalate-only safety rules
- Automated resident clarification loop
- Eligible vendor candidate matching

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03 TARGET STATE — RESULT

- Emergencies wake on-call staff instantly
- Prepared vendor draft ready for 1-tap send
- Routine tickets synced clean into PMS
- 100% human control over vendor contact

Note on conceptual scope: *This document outlines a potential implementation tailored for Highland Property Group. Specific approval thresholds, communication channels, and software connectors would be aligned with your existing operating procedures prior to setup.*

LOGICAL PROGRESSION & DECISION GATES

How The System Operates Inside Your Business

A visual walkthrough showing how inbound reports are captured, safely evaluated, clarified, and handed to your team for approval.

VISUAL COMPONENT GRAMMAR: External Input · Automation Action · AI Interpretation · Human Checkpoint · Executed Outcome

STAGE 01 — Multi-Channel Intake

Resident Messages — Trigger
Email, Portal Form, or SMS from tenant arrives 24/7.

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Normalize & Dedupe — Core
Strips signatures, computes body hash, filters 24h repeats.

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Preliminary Buffer — Guard
Vague greetings held until a concrete symptom is stated.

STAGE 02 — Triage & Safety

AI Interpretation — Cognitive
Extracts Trade, Room, Fixture, Access info, and preliminary Urgency.

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Binding Safety Rules — Escalate-Only
Deterministic regex checks 10 hazard types (gas, flood, fire). Can raise, never lower.

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Durable Persistence — Database
Allocates permanent reference (e.g. MR-0012) & audit history.

STAGE 03 — Routing & Filters

MISSING INFO **Tenant Clarification:** Asks resident for missing access/pet/symptom details. Manager unalerted.

SENSITIVE CASE **Human Review Only:** Legal threats, accommodation, or habitability disputes skip auto-drafts.

COMPLETE & SAFE **Vendor Match:** Filters active vendor roster by property & trade; prepares natural dispatch draft.

STAGE 04 — Human Checkpoint

Manager Alert & Choices — Approval Point
Mobile/Slack notification displays original words, AI summary, trade, and buttons for matched vendors.

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One-Send Lock — Idempotency
Database lock guarantees vendor is emailed at most once, even if tapped twice.

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Approved Dispatch — Action
Email sent to chosen contractor with job scope & property details.

STAGE 05 — Sync & Status

PMS Synchronization — PM System
Work order notes, assigned vendor, and timestamp synced to AppFolio/Buildium.

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Factual Resident Update — Confirmed Send
Tenant notified *only after* vendor send succeeds. Never promises unconfirmed arrival times.

Unclear / Garbled Message
Never silently dropped. Stored with full raw text and assigned to coordinator review queue with reason flagged.

Vendor Email Delivery Fails
System records failure, alerts manager for manual re-try, and explicitly blocks resident "help is coming" update.

Unanswered Emergency
Automated 15-minute scheduled check monitors unacknowledged emergencies and escalates to secondary on-call roster.

ENTERPRISE COMPATIBILITY & BOUNDARIES

How It Integrates With Your Existing Operation

The system acts as an intelligent operating layer around your software. You do not replace your PMS or retrain your team.

LAYER 01 // CHANNELS
Existing Inbound

- Resident Web Portal
- Maintenance Inbox (Gmail / M365)
- Resident SMS Line
- After-Hours Voicemail Line

▼ Non-disruptive webhook / mailbox listener

LAYER 02 // OVERLAY
Sambhav Builds Engine

- Message Normalizer
- Strict Safety Regex Engine
- AI Structured Classifier
- Clarification Loop
- Approval Lock

▼ Controlled two-way data sync & audit events

LAYER 03 // BUSINESS RECORD
AppFolio / Existing PM System

- Authoritative Leases & Units
- Work Orders & Tasks
- Approved Vendor Directory
- Accounting & Invoices

▼ Real-time notifications & mobile approvals

LAYER 04 // STAKEHOLDERS
People & Partners

- Maintenance Coordinators
- On-Call Property Managers
- Qualified Trade Vendors
- Rental Property Owners

Zero Software Changes

Your property management system remains the authoritative source of truth. No data migration or platform replacement required.

Zero Resident Friction

Residents use the email addresses, web forms, and phone numbers they already know. No new mobile apps to download.

Zero Unapproved Dispatches

The system is strictly advisory. No vendor is ever contacted and no financial commitment made without explicit human sign-off.

Items We Validate Together Prior to Implementation

Feasibility Checklist

01 PM Software Connector — Verify
Confirm API access or email-parser integration for AppFolio / Existing PM System.

02 Spending Thresholds — Verify
Review your standard owner approval limits (typically $250–$500) and emergency override policies.

03 Alert Channels — Verify
Choose manager notification destination: Slack, Microsoft Teams, SMS, WhatsApp, or email.

04 Vendor Trade Rosters — Verify
Map preferred plumbing, HVAC, electrical, and general maintenance vendors by property zone.

05 Sensitive Boundaries — Verify
Identify legal, accommodation, and Fair Housing keywords reserved exclusively for human handling.

06 Quiet Hours & On-Call — Verify
Define after-hours escalation rules and on-call supervisor phone rotation schedules.

LOW-FRICTION NEXT STEP

15-Minute Technical Feasibility Alignment

If this concept addresses a tangible bottleneck in your operation, we spend 15 minutes reviewing your current intake channels and PM software setup to confirm practical feasibility. Zero sales pitch, no commitment.

Sambhav Jain • AI & Workflow Automation Engineer
sambhav.builds@gmail.com • Direct Message on LinkedIn